InternNova Week 6 - Final Data Analytics Project

Online Retail Sales & Customer Analytics

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End-to-End Data Analytics Capstone

Project Scope	Data Cleaning, EDA, Python Visualization, Power BI, Git & GitHub
Dataset	UCI Online Retail transaction dataset
Final Dataset Rows	524,878 cleaned sales rows
Total Revenue	10,642,110.80
Total Orders	19,960

1. Executive Summary

This capstone presents an end-to-end analysis of an online retail transaction dataset using Python, pandas, matplotlib, Power BI, Git and GitHub. The objective was to clean raw transactional data, explore customer and sales behavior, create meaningful visualizations, build an interactive business dashboard, and translate analytical findings into practical recommendations.

The cleaned sales dataset contains **524,878** valid sales rows. The analysis identified total revenue of **10,642,110.80** across **19,960** distinct orders. The United Kingdom dominates sales value, while revenue accelerates strongly from September through November 2011, with November reaching approximately **1.50 million**.

2. Problem Statement

The project aims to understand how sales, products, countries, customers and transaction timing influence retail performance. The business goal is to identify high-value markets, high-performing line items, seasonal trends and potential operational actions that can improve inventory planning, marketing focus and customer targeting.

3. Dataset Description

Field	Description
InvoiceNo	Transaction / invoice identifier
StockCode	Product stock code
Description	Product description
Quantity	Units purchased
InvoiceDate	Transaction date and time
UnitPrice	Price per unit
CustomerID	Customer identifier
Country	Customer country

Raw dataset size: **541,909 rows and 8 columns**. Missing values were present mainly in CustomerID and Description, and 5,268 exact duplicate rows were identified.

4. Data Cleaning and Preparation

The raw data was cleaned in Python with pandas. A separate sales dataset was maintained for overall sales analysis, while customer-level analysis excluded rows without CustomerID.

- Removed **5,268 exact duplicate rows**.
- Excluded rows with Quantity <= 0 because these represented cancellations, damage, stock adjustments or other non-sales activity.
- Excluded rows with UnitPrice <= 0.
- Excluded rows with missing Description from sales analysis.
- Created a Revenue feature as **Quantity x UnitPrice**.
- Created Year, Month, Month_Name, Quarter, Day and Hour features from InvoiceDate.
- Preserved rows without CustomerID for total sales analysis, but excluded them from customer-specific analysis.

Metric	Result
Rows after duplicate removal	536,641
Final cleaned sales rows	524,878
Customer-analysis rows	392,692
Negative quantity rows before sales filtering	10,587
Cancelled invoice rows after deduplication	9,251
Quantity outliers (IQR)	27,111
Revenue outliers (IQR)	42,624

5. Exploratory Data Analysis

The EDA focused on overall sales performance, monthly trends, country contribution, product/line-item contribution, outliers and the relationship between Quantity, UnitPrice and Revenue.

KPI	Value
Total Revenue	10,642,110.80
Distinct Orders	19,960
Revenue vs Quantity correlation	0.907
Revenue vs UnitPrice correlation	0.137
Highest revenue month	November 2011 (~1.50M)
Lowest full month	February 2011 (~0.52M)

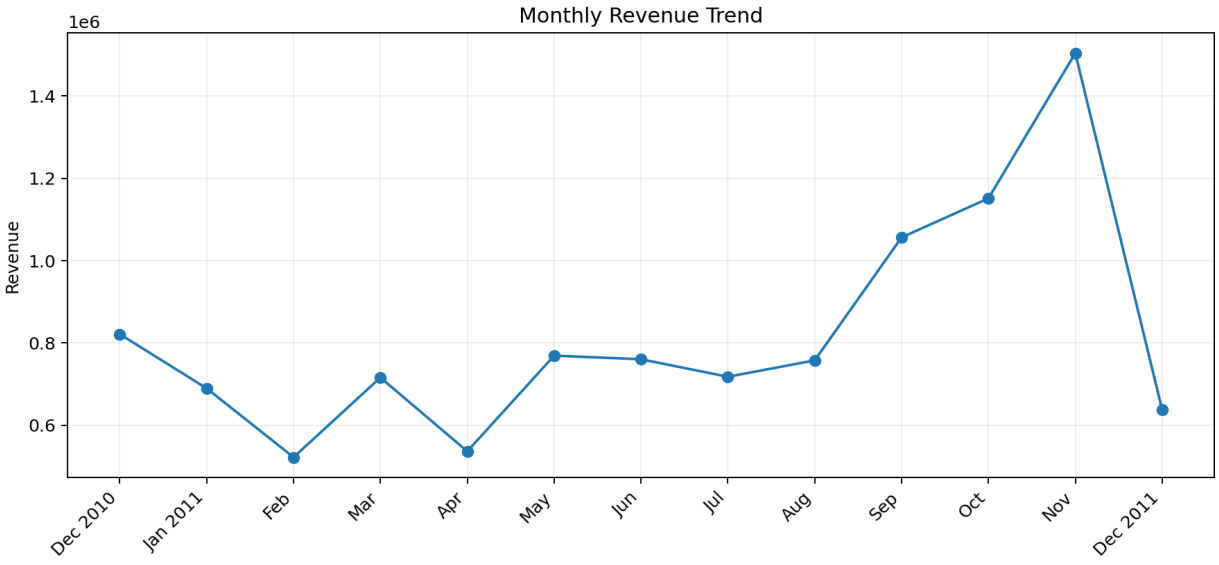


Figure 1. Monthly revenue shows a strong seasonal ramp from September to November. December 2011 is incomplete and should not be compared as a full month.

6. Country and Product Analysis

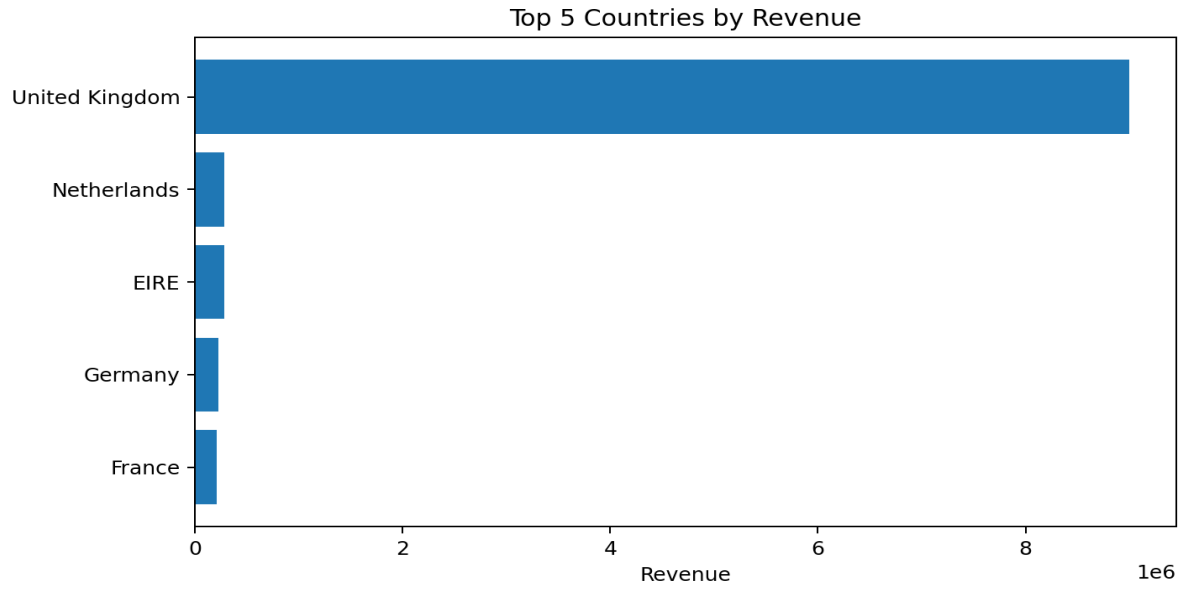


Figure 2. The United Kingdom contributes approximately 9.00M in revenue, far exceeding other markets.

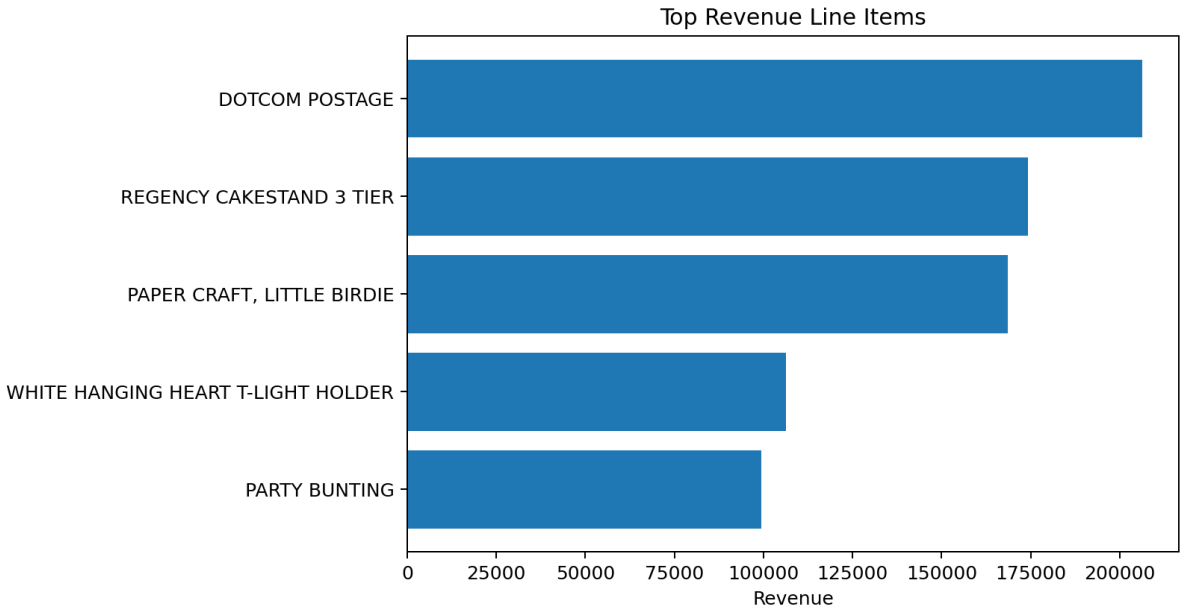


Figure 3. Leading revenue line items include DOTCOM POSTAGE, REGENCY CAKESTAND 3 TIER and PAPER CRAFT, LITTLE BIRDIE.

7. Python Visualizations

Six final Python visualizations were produced and stored in the GitHub Visualizations folder:

- Monthly Revenue Trend
- Top 10 Countries by Revenue
- Top 10 Products by Revenue
- Revenue Distribution
- Quantity vs Revenue
- Revenue by Hour

These visualizations support time-series analysis, market ranking, product performance, distribution analysis and relationship analysis.

8. Power BI Dashboard

A Power BI dashboard was created with KPI cards, country and month slicers, monthly revenue analysis, top products, top countries and hourly revenue analysis. The PBIX file is included in the GitHub repository.

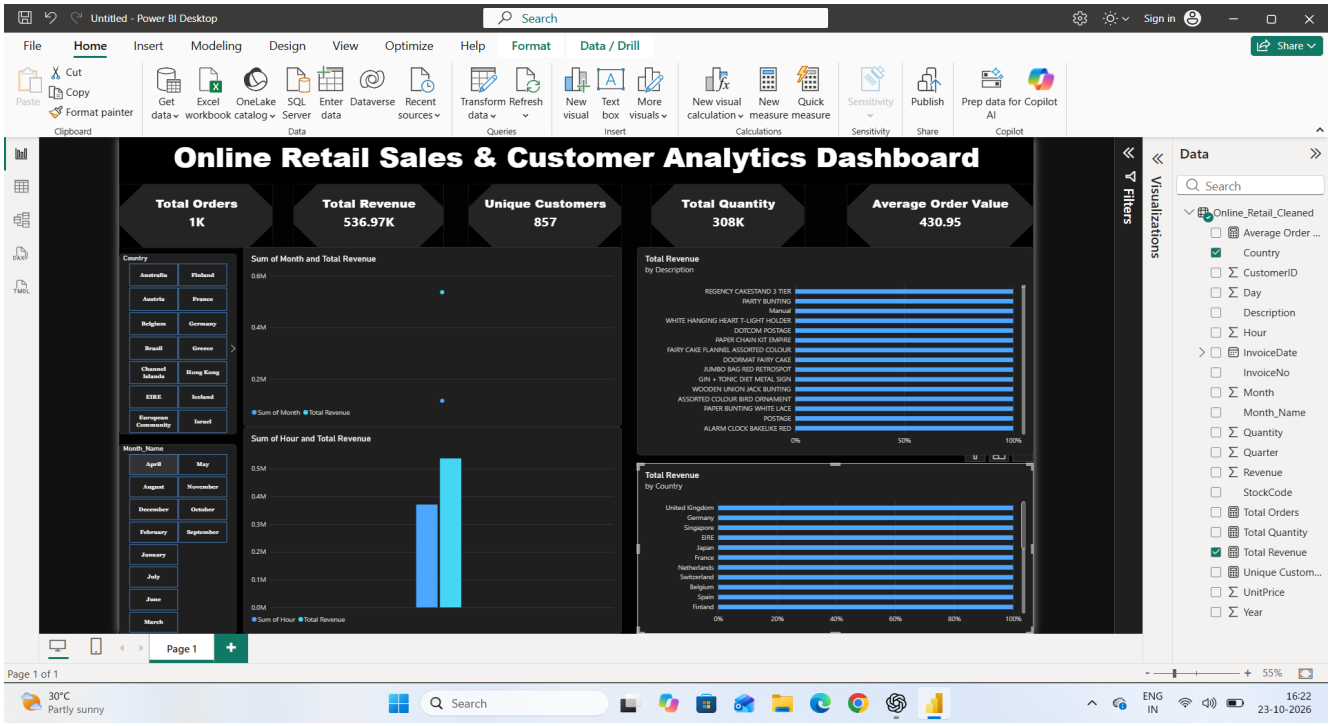


Figure 4. Power BI dashboard screenshot. The displayed KPI values in this screenshot reflect the active filtered state of the dashboard, while the overall project totals are reported from the full cleaned dataset.

9. Business Insights and Recommendations

Key Insights

- The United Kingdom is the dominant market, contributing roughly 9.00M in revenue, while Netherlands, EIRE, Germany and France form the next tier of markets.
- Revenue rises sharply toward the end of 2011. September, October and November are the strongest consecutive months, with November the highest at about 1.50M.
- Quantity has a strong positive relationship with Revenue (correlation ~0.907), while UnitPrice has a much weaker relationship (~0.137).
- High-value line items include DOTCOM POSTAGE, REGENCY CAKESTAND 3 TIER and PAPER CRAFT, LITTLE BIRDIE.
- The dataset contains substantial quantity and revenue outliers, indicating bulk or unusually large transactions that should be monitored separately.

Recommendations

- Increase inventory readiness and marketing activity before the September-November peak period.
- Protect the United Kingdom customer base while selectively expanding activity in Netherlands, EIRE, Germany and France.
- Develop offers for high-volume customers and bulk buyers, such as loyalty programs, volume discounts or targeted bundles.
- Separate operational/administrative line items such as postage from merchandise product rankings when making product-level decisions.

10. Git and GitHub Workflow

Git and GitHub were used to maintain project versions and publish the final capstone. The workflow included repository creation, cloning, staging, meaningful commits, pulling remote changes, resolving local/remote synchronization, and pushing the completed project.

- git clone
- git status
- git add
- git commit
- git pull --rebase
- git push

The final repository contains the raw dataset, cleaned dataset, Jupyter notebook, Power BI file, presentation, visualizations, screenshots and README documentation.

11. Tools and Technologies

Tool	Use
Python	Data cleaning, EDA, analysis and visualization
pandas	Data manipulation and feature engineering
matplotlib	Charts and visual analysis
Power BI	Interactive dashboard and KPI reporting
Git	Local version control
GitHub	Repository hosting and project presentation
VS Code / Jupyter	Notebook development environment

12. Conclusion

This project demonstrates a complete data analytics workflow from raw data to business-ready insights. The analysis combined data quality checks, transformation, exploratory analysis, visual storytelling, dashboard development and version-controlled documentation. The results show that market concentration, seasonal demand and transaction volume are major drivers of performance in this retail dataset.

The capstone also demonstrates practical use of Python, Power BI and GitHub in a portfolio-style project suitable for demonstrating end-to-end analytics capability.

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